

Part 2 - The hearing process

Note: In "Hearing process" mode, the "Export" button opens a selection window. There you can specifically export the simulation window, the "Cochlea (with legend)" or the "Activation of auditory sensory cells". In the sequence task, you can also save the exported solution after solving it correctly.

For a sound to be perceived, the vibration of the air must be passed on and transformed through different structures of the ear in sequence. The simulation shows this pathway in simplified form in the simulation window and supplements it with representations of the cochlea and the auditory sensory cells.

Task 1: Structure of the ear

Open Task 1 "Structure of the ear". Match the terms to the structures in the figure correctly and then export the result.

Structure of the ear
Insert here

Task: Now draw the areas outer ear, middle ear and inner ear into the exported image. Separate the areas with vertical lines.

Task 2: Sequence of the hearing process

Open Task 2 "Put the hearing process in the correct order". Arrange the blocks correctly, check your solution and then export the result.

Sequence of the hearing process
Insert here

How is the signal transmitted in the outer, middle and inner ear?

Task 3: Cochlea and fluid movement

In the export window, choose "Cochlea (with legend)". Use the representation to explain how the movement in the cochlea arises, is transmitted and finally ends.

Cochlea
Insert here

Task 4: Activation of the auditory sensory cells

In the export window, choose "Activation of the auditory sensory cells". Describe how the electrical signal is produced.

Activation of auditory sensory cells
Insert here
